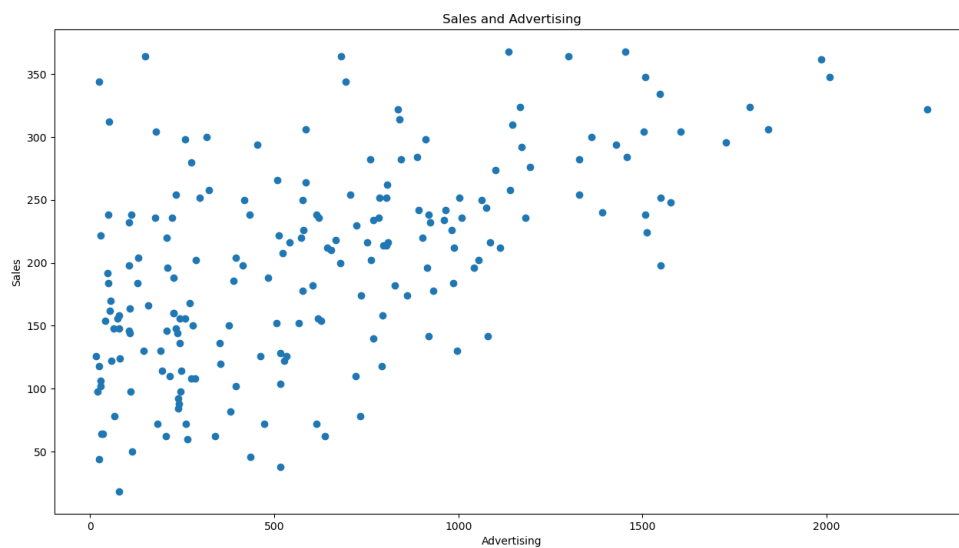


Report:

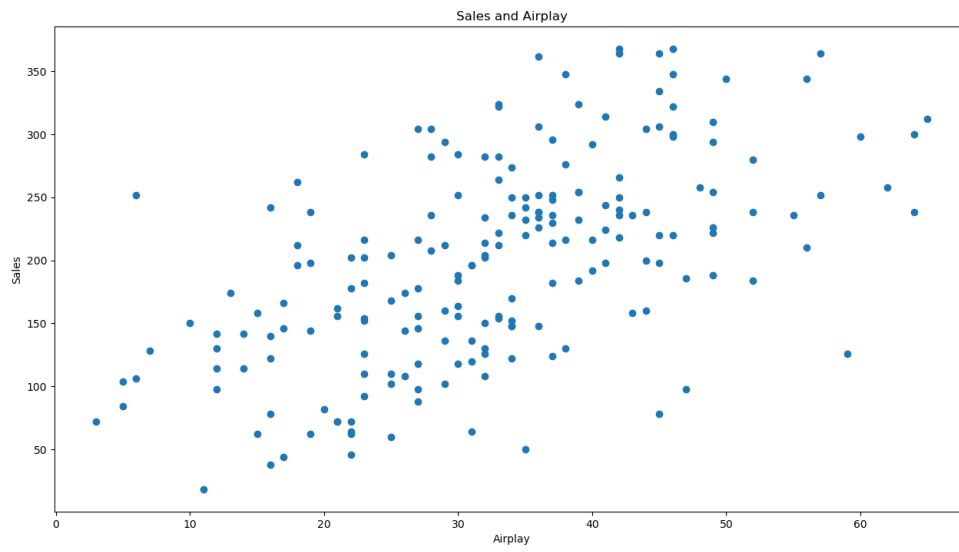
Exercise 5 – Regression and Multiple Regression

Visualization:

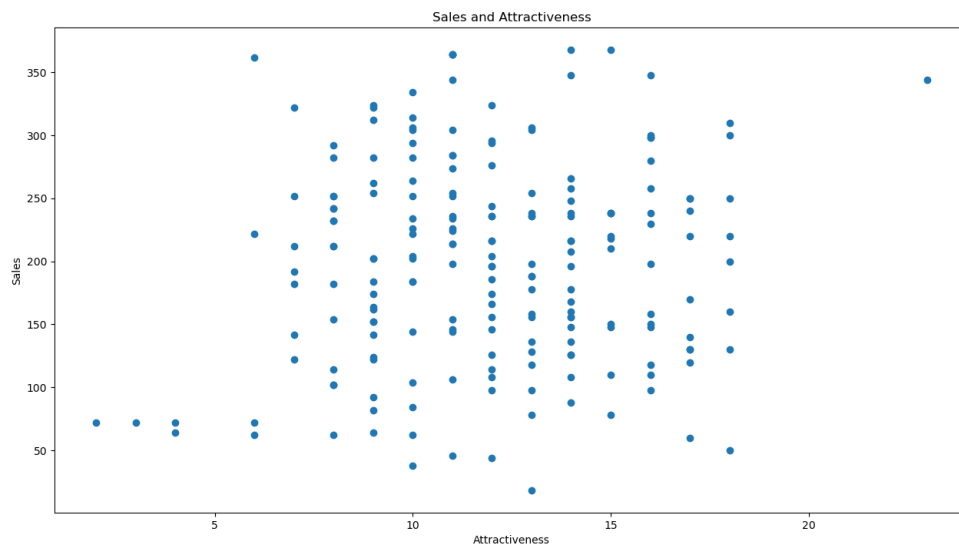
1. Sales and advertising



2. Sales and airplay



3. Sales and attractiveness



Linear Regression:

4. Conduct a linear regression to construct a linear model between Sales and adverts and write down the F-statistic and P-value.

Linear model: $y = 0.09551716452262712 * \text{advertising} + 139.59312007800236$

F-statistic: 98.04

P-value: 4.946499173591511e-19

5. Discuss what these values (F-statistic and P-value) describe about our linear regression model. Is it good? Bad? Can't say?

A: The F-statistic value of 98.04 indicates that the relationship between advertising and sales is strong and it is not likely to be due to random chance. The P-value supports this since its value is very small. These values are good because it shows that there is a strong relationship between advertising and sales.

Model Coefficients:

6. What is the intercept value and coefficient (adverts) value of your linear regression model?

Intercept value: 139.59

Coefficient value: 0.0955

7. Using the intercept value and coefficient of your linear model, please calculate how many records will be sold if we spent \$135,000 on advertising the latest album "Dear Agony" by Breaking Benjamin.

For \$135,000 spent on advertising: 13034 records sold

Multiple Regression:

8. Conduct a multiple regression to construct a model between Sales and the predictors (adverts, airplay, attract) and report the F-statistic and P-value.

F-Statistic: 105.23

P-value: 1.2726224041901978e-40

9. We know that the R-squared value can be used to evaluate the overall fit of a linear model. Also, higher R-squared values are better if their p-values are less than 0.05.

Based on this, discuss which one of the two models that you constructed is better?

- Model 1: the linear model between Sales and advertising you constructed in (4).
- Model 2: the multiple regression model between outcome: Sales and the predictors (ad-

vertising, airplay, attractiveness) that you constructed in (8).

A: Model 1's r^2 value is 0.331, which tells us that advertising explains 33.1% of the variance in sales, and that the relationship is statistically significant. Also, the P-value for model 1 is very small, which is much less than 0.05. This also indicates a strong relationship. For model 2, the r^2 value is 0.617, which tells us that the combining of the variables advertising, airplay, and attractiveness explain 61.7% of the variance in sales, and that the model is highly significant. Overall, model 2 is better because it has a higher r^2 value than model 1. Both models are statistically significant, but model 2 provides a better prediction because it uses multiple predictors.